

Course	Fundamentals of Fog & Edge	Date	10-03-2025
Name	Computing		
Course Code	BCSE313L	SLOT	F1 & F2
		Class No.	VL2024250501570/1575

Tutorial Sheet 4 and 5

Problem Statement 1

Why is minimizing latency considered a major driving force behind fog computing, and how does it differ from traditional cloud computing? Justify your answer.

Problem Statement 2

How does energy consumption in fog computing systems differ across layers (end devices, edge resources, and the cloud), and why is optimizing energy usage critical for each layer? Justify your reasoning.

Problem Statement 3

Why is Pareto-optimality an important concept in fog computing optimization, and how does it address the challenge of conflicting objectives like latency and energy consumption? Justify your answer.

Problem Statement 4

How does the selection of surrogate devices in fog and edge computing impact system performance, and why is energy-aware selection considered a trade-off? Justify your reasoning.

Problem Statement 5

A real-time video streaming service is being deployed using a fog computing architecture. The system consists of a central cloud data center, multiple geographically distributed edge resources (fog nodes), and end-user devices such as smartphones and smart TVs. The goal is to optimize the system for low latency while minimizing energy consumption in both the cloud and edge resources. In this scenario, how would you balance the trade-off between minimizing latency and reducing energy consumption when deciding where to process video streams (cloud vs. edge resources)? Justify your reasoning based on the hierarchical three-layer model of fog computing.

Problem Statement 6

A mobile application allows users to perform compute-intensive tasks (e.g., image recognition or augmented reality) on their smartphones. To reduce the energy consumption of the smartphones and improve performance, the application offloads

some tasks to nearby edge resources. However, the edge resources have limited capacity, and the network bandwidth between the smartphones and edge resources is constrained. How would you decide which tasks should be offloaded to the edge resources and which tasks should remain on the smartphones? Justify your reasoning based on the optimization objectives of energy consumption, execution time, and network bandwidth usage.

Problem Statement 7

A city has deployed a real-time traffic monitoring system using fog and edge computing infrastructure. The system uses IoT sensors installed at traffic intersections to collect data about vehicle movement, congestion, and accidents. The collected data is processed by nearby edge devices (fog nodes) to generate real-time updates for traffic management systems and emergency response teams. However, the system must handle dynamic changes in traffic patterns, device mobility, and network connectivity. In this scenario, how would context-aware adaptive design in the middleware help improve the performance of the traffic monitoring system? Justify your reasoning based on the challenges posed by dynamic user environments and resource constraints.

Problem Statement 8

A mobile crowdsensing application is designed to gather environmental data (e.g., air quality, noise levels) from volunteer users' smartphones. The application uses fog nodes to process and aggregate the collected data before sending it to the cloud for further analysis. However, the application faces challenges such as limited battery life on smartphones, varying user participation, and the need for secure data transmission. How would energy-aware surrogate selection and security mechanisms in the middleware address the challenges faced by the mobile crowdsensing application? Justify your reasoning based on the trade-offs involved in selecting participating devices and ensuring data privacy.

Problem Statement 9

A busy urban area has deployed a Smart Parking System using IoV technologies to address parking challenges. The system enables vehicles to communicate with roadside units (RSUs) and cloud servers to locate available parking spaces in real time. Drivers can access the system through their smartphones or in-vehicle infotainment systems. The system also integrates payment gateways, allowing users to reserve and pay for parking slots seamlessly. How do the different types of communication in IoV (V2V, V2I, V2R, V2P, V2S) contribute to the functionality and efficiency of the smart parking system? Justify your reasoning by explaining the role of each communication type in this scenario.

Problem Statement 10

A city has implemented an Emergency Response System using the Internet of Vehicles (IoV) to improve road safety and reduce response times during accidents. The system integrates vehicles, roadside units (RSUs), and cloud infrastructure to enable real-time communication. When an accident occurs, nearby vehicles detect the event using onboard sensors and cameras, and the data is transmitted to RSUs and the cloud for further processing. The system then alerts emergency services, nearby vehicles, and pedestrians via smartphone notifications. In this scenario, how does the multi-layered architecture of IoV (Perception, Coordination, AI, Application, and Business Layers) contribute to the effectiveness of the emergency response system? Justify your reasoning by explaining the role of each layer in handling the accident detection and response process.